

SANDY LAKE A, ON, CanadaWMO: **715870**Lat: **53.063N**Lon: **93.344W**Elev: **290**StdP: **97.89**Time Zone: **-6.00 (NAC)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-32.4	-39.4	0.1	-35.0	-36.4	0.1	-32.2	8.3	-16.5	7.4	-17.7	1.4	300	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	27.7	18.2	26.2	17.4	24.8	16.7	20.1	25.4	19.0	23.8	18.0	22.6	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.7	22.1	17.2	12.7	21.1	16.2	11.9	20.4	59.1	25.6	55.3	23.9	52.2	22.6	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.0	7.0	6.2	-38.8	32.1	2.2	2.1	-40.4	33.7	-41.7	34.9	-43.0	36.2	-44.6	37.7
			-39.3	22.4	1.9	1.1	-40.7	23.2	-41.8	23.8	-42.8	24.4	-44.2	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-19.3	-18.2	-9.8	-1.0	7.9	14.9	17.9	16.2	10.6	3.4	-7.7	-16.9
	DBStd	14.68	7.78	7.16	7.39	7.08	5.50	4.17	3.53	3.56	4.91	4.87	7.20	7.98
	HDD10.0	4390	908	789	615	332	108	7	0	1	51	214	531	834
	HDD18.3	6792	1166	1022	873	581	326	117	51	84	237	463	781	1093
	CDD10.0	719	0	0	0	1	43	155	246	195	69	9	1	0
	CDD18.3	77	0	0	0	0	2	14	38	19	4	0	0	0
	CDH23.3	658	0	0	0	0	39	132	317	137	33	1	0	0
	CDH26.7	112	0	0	0	0	6	13	69	19	6	0	0	0
	WSAvg	2.9	2.7	2.7	3.0	3.0	3.2	2.9	2.8	2.8	2.7	3.1	2.9	2.5
	PrecAvg	553	22	18	24	25	46	69	91	84	65	51	33	23
(15) Precipitation	PrecMax	733	52	39	52	50	89	138	146	165	114	148	52	50
	PrecMin	456	8	6	12	3	14	29	39	29	15	15	15	8
	PrecStd	69	10	8	11	12	18	23	26	30	27	25	10	10
	PrecStd	69	10	8	11	12	18	23	26	30	27	25	10	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	1.9	12.9	18.1	27.3	28.2	31.7	29.1	26.6	20.0	11.4	1.7
		MCWB	0.3	-0.1	8.4	8.5	14.6	17.9	21.1	20.1	17.2	13.4	9.0	-0.1
	2%	DB	-1.3	-1.0	7.3	15.1	23.6	26.5	28.4	26.5	23.7	16.1	6.7	-1.4
		MCWB	-1.7	-2.2	3.1	7.3	12.9	16.8	19.1	18.3	17.2	11.3	5.3	-2.2
	5%	DB	-3.9	-4.0	3.4	12.6	20.5	24.9	26.6	24.8	21.1	13.4	3.5	-3.2
		MCWB	-4.6	-5.1	0.2	6.0	11.7	15.9	18.0	17.4	15.8	10.4	2.4	-3.7
	10%	DB	-7.4	-6.7	1.0	9.8	17.5	23.1	24.9	23.0	18.5	10.7	1.1	-5.5
		MCWB	-7.9	-7.7	-0.7	4.4	10.9	15.1	17.2	16.9	14.2	8.5	0.4	-5.9
	0.4%	WB	0.4	0.9	7.9	9.6	16.8	20.9	22.4	20.9	19.9	14.2	8.5	0.1
		MCDB	0.7	1.8	13.5	16.4	23.9	25.3	29.1	26.9	24.8	17.9	11.0	1.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-1.6	-2.3	3.1	7.8	14.4	18.5	20.6	19.4	18.0	12.4	5.2	-2.0
		MCDB	-1.4	-1.3	6.4	14.4	20.6	23.3	26.0	24.3	22.5	15.3	6.5	-1.5
	5%	WB	-4.6	-5.1	0.7	6.3	12.7	17.3	19.2	18.5	16.2	10.4	2.4	-3.7
		MCDB	-4.0	-4.1	2.5	11.6	19.0	22.6	24.5	22.9	19.8	12.8	3.4	-3.2
	10%	WB	-7.9	-7.7	-0.8	4.6	11.2	16.2	18.3	17.5	14.7	8.5	0.4	-5.9
		MCDB	-7.5	-6.6	1.5	9.4	17.1	21.4	23.0	21.8	18.4	11.0	1.0	-5.4
	5% DB	MDBR	9.2	11.5	12.9	12.9	11.9	11.2	10.8	10.1	9.1	6.4	6.5	7.6
		MCDBR	9.1	12.0	13.3	15.3	16.0	13.6	13.0	12.6	12.2	10.7	8.0	6.7
		MCWBR	8.8	10.5	9.8	9.0	7.9	6.2	5.7	6.1	6.7	6.8	6.4	6.1
		MCWBR	9.1	11.4	11.1	14.3	14.2	11.4	11.1	10.8	10.8	9.7	7.7	6.6
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.8	10.2	8.7	8.7	7.6	6.1	5.7	6.1	6.7	6.6	6.4	6.0
		MCWBR	8.8	10.2	8.7	8.7	7.6	6.1	5.7	6.1	6.7	6.6	6.4	6.0
	Clear-Sky Solar Irradiance	taub	0.239	0.255	0.275	0.319	0.343	0.371	0.386	0.364	0.328	0.291	0.263	0.229
		taud	2.285	2.280	2.318	2.392	2.455	2.396	2.392	2.453	2.524	2.585	2.426	2.293
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	770	868	922	917	907	880	858	861	857	829	740	713
		Edh at Noon	58	81	98	105	104	112	110	98	81	62	52	48
(44) All-Sky Solar Radiation	RadStd	RadAvg	1.05	2.25	3.69	5.20	5.28	5.55	5.42	4.45	3.05	1.70	0.85	0.66
		RadStd	0.11	0.15	0.29	0.37	0.43	0.52	0.42	0.35	0.41	0.19	0.09	0.10

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days
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